Snowstorm

Snowstorm is a white noise drone/ring modulated distortion pedal.

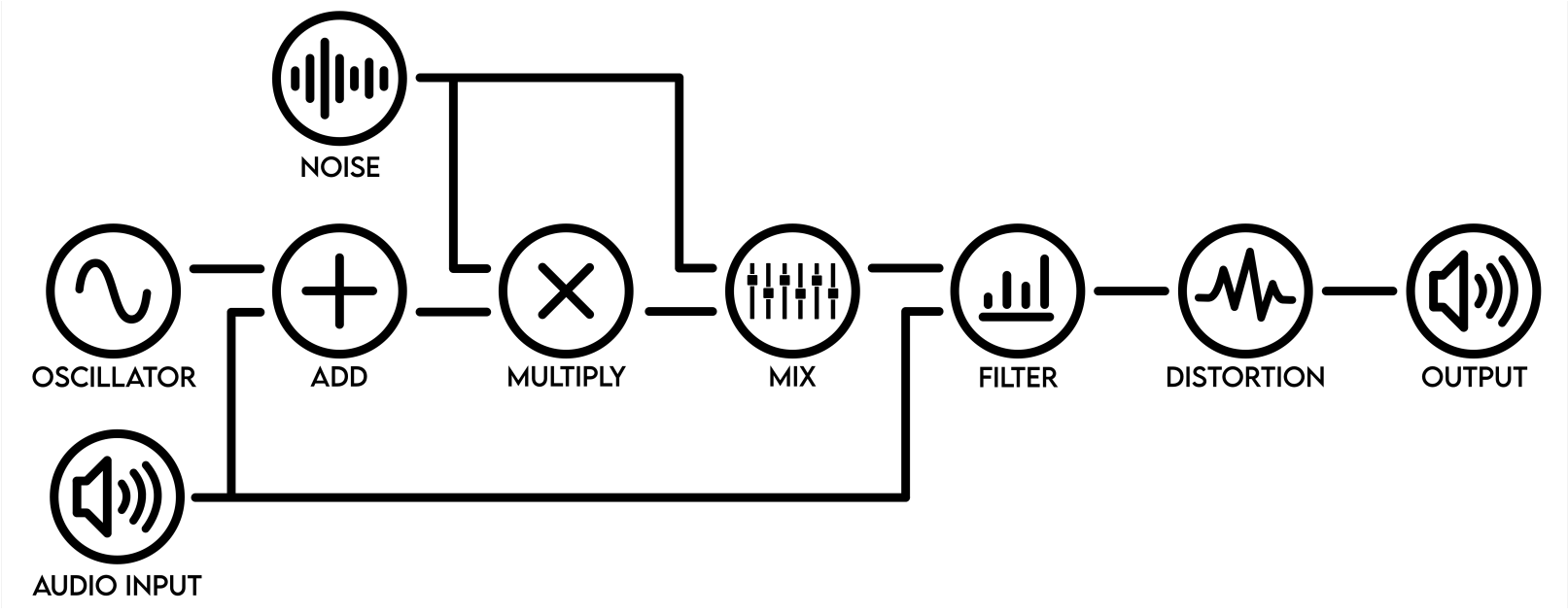
In its droning state this pedal works by ring modulating a noise source with a droning oscillator. A dry copy of the noise is then added back in with the ring modulated output. The free running oscillator can either be a sine wave or a rising saw wave.

You can also use the audio input as a substitute layer for the oscillator, or a second carrier source that mixes with the oscillator to add into the ring modulation section.

This signal then gets sent into either a lowpass, highpass or comb filter followed by one of six distortion algorithms.

The density of the noise will greatly affect how filthy the signal gets as the carrier source gets lost in the snowstorm of noise. Perfect for walls of noise and bass, or for gnarly industrial distortion that sounds like disintegrating speakers.

This pedal is capable of outputting sub bass frequencies with the oscillator, as well as outputting very loud volumes with the white noise. So please be mindful of what speaker setup you use if engaging with this aspect of the pedal, as damage could occur if pushed in a way it was not originally designed for. Full range full frequency monitors and/or bass amps handle this great!



Hardware Functions

Frequency (Knob 1) - Frequency controls the frequency of the internal sine wave or saw wave oscillator. This oscillator ranges from C-2 - C8.

Density (Knob 2) - Density controls the density of the noise. When counterclockwise it will produce a light sputtering of noise. When fully clockwise it will produce a dense wall of noise.

Distortion (Knob 3) - Distortion controls the amount of distortion generated by one of the six available distortion algorithms. Changing dipswitch 4 will swap the original three algorithms out for a different set of three.

Cutoff (Knob 4) - Cutoff controls the frequency cutoff for the three available filters. The filters range from 20Hz - 20,000Hz

Resonance (Knob 5) - Resonance controls the resonance for the three available filters. When fully counterclockwise it will produce a smooth/tame filter sweep. When fully clockwise it will transform the filters into a very sharp sweep, as it highly emphasizes the cutoff's value.

Volume (Knob 6) - Volume controls the output volume. Use this knob to prevent any clipping, or don't. As it can be leaned into for even more gnarly distortion. It may help to have a limiter after Snowstorm if you decide to lean into this, as this pedal can get VERY loud.

Noise (Toggle 1) - Choose between either white, pink or brown noise.

Distortion (Toggle 2) - Choose between one of three distortion algorithms.

Filter (Toggle 3) - Choose between a lowpass filter, highpass filter or comb filter.

Dipswitch 1 - Density or Cutoff control for the expression pedal

Dipswitch 2 - Audio input mix. 0% or 50% wet mix.

Dipswitch 3 - Sine wave or saw wave

Dipswitch 4 - Distortion bank 1 or 2

Stomp Switch Left - This enables or disables the internal oscillator.

Stomp Switch Right - This bypasses the effect entirely.

Expression Pedal - This is mapped to either the density or cutoff function. You can attenuate the expression pedal's range with the second dipswitch. Use a ¼" TRS to ⅛" TRS cable for this function. It MUST be TRS and not TS for the expression pedal to work properly.

MIDI - All knobs, toggle switches, dip switches and stomp switches are mapped to MIDI CC controls. Use a MIDI DIN to ⅛" TRS cable to access this function. It MUST be TRS and not TS for the MIDI to work properly.

When MIDI CC data is sent the physical knobs position is then ignored in favour of the MIDI CC data. However, if the MIDI CC data remains static moving the knob, toggle switch, dip switch or stop switch will reengage the value as dictated by the hardware's position.

Snowstorm is set to MIDI channel 1 by default.
If you would prefer a specific MIDI channel and CC setup, please feel free to email me and I'll send over a custom tailored MIDI CC bin file.

MIDI CC Mapping

Snowstorm is set to MIDI Channel 1 by default.

Knob 1 - CC1 - Frequency
Knob 2 - CC2 - Density
Knob 3 - CC3 - Distortion
Knob 4 - CC4 - Cutoff
Knob 5 - CC5 - Resonance
Knob 6 - CC6 - Volume

Toggle 1 - CC7 - Noise
Toggle 2 - CC8 - Distortion
Toggle 3 - CC9 - Filter

Dipswitch 1 - CC10 - Expression pedal: density or cutoff
Dipswitch 2 - CC11 - Expression pedal attenuation range
Dipswitch 3 - CC12 - Sine or saw wave
Dipswitch 4 - CC13 - Distortion bank 1 or 2

Stomp Switch Left - CC14 - Engage oscillator
Stomp Switch Right - CC15 - Bypass